

Yashas Kadambi

M.S. Computer Science, UC San Diego | Prior Teaching Assistant | Software Engineer

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TEACHING PROFILE

Prior TA experience in Image Processing and Computer Vision, Data Analytics, and Deep Learning Theory and Practices, supporting 656 students across three courses. Designed assignments, labs, answer keys, solution notebooks, and instructional recap videos, and built grading workflows for nearly 800 undergraduate assignment, lab, and competition entries, plus three graduate-course assignments. Also brings 3+ years of software engineering experience at Cisco and two peer-reviewed publications.

EDUCATION

University of California San Diego

M.S. in Computer Science | San Diego, California

Expected 2027

PES University

B.Tech. in Computer Science and Engineering | TEC-evaluated U.S. GPA: 3.78/4.00 (124.5 semester credits) | MRD Scholarship

Aug 2019 - May 2023

TEACHING EXPERIENCE

PES University | Teaching Assistant

Department of Computer Science and Engineering | Bengaluru, India

Jul 2022 - May 2023

Image Processing and Computer Vision | 122 students

Instructor: Prof. Gowri Srinivasa

Dec 2022 - May 2023

- Designed one MATLAB-based steganography assignment, one hands-on lab, teaching materials for one course unit, and a complete answer key covering image sampling, intensity transformations, histogram equalization, ROI masking, and LSB watermarking.
- Graded approximately 250 submissions within one week; answered approximately 50 student questions and held five 2-hour office-hour sessions.
- Built a reusable grading macro that automated repetitive checks and reduced assignment-evaluation time by 50%.
- Tech stack:** MATLAB and Image Processing Toolbox.

Data Analytics | 494 students

Instructor: Prof. Gowri Srinivasa

Aug 2022 - Dec 2022

- Designed a Brooklyn Nine-Nine-themed assignment on Markov chains, absorbing states, stationary distributions, and A/B testing; created and hosted a public [Kaggle team competition](#); and authored unit materials, a complete answer key/solution notebook, and instructional recap videos.
- Manually graded approximately 370 assignment submissions within one week; answered approximately 150 questions through six 2-hour office-hour sessions and course forums.
- Built the competition's automated grader and manually verified submissions from 178 teams, returning consistent results within one day.
- Tech stack/platforms:** Python, NumPy, pandas, Matplotlib, scikit-learn, Jupyter/Google Colab, and Kaggle.

Deep Learning Theory and Practices | ~40 students

Instructors: Prof. Preet Kanwal and Prof. Uma P

Jul 2022 - Dec 2022

- Selected as an undergraduate TA for the ~40-student graduate course Deep Learning Theory and Practices; authored three assignments, hands-on labs, and teaching materials on autoencoders and generative adversarial networks.
- Developed an MNIST autoencoder notebook covering encoder-decoder construction, custom training, image reconstruction, and TensorBoard visualization.
- Built a web-based submission and autograding workflow that returned instant validation for all three assignments; supported students through two dedicated office-hour sessions and hands-on debugging.
- Tech stack:** Python, NumPy, pandas, TensorFlow/Keras, PyTorch, and TensorBoard.

TEACHING AND TECHNICAL SKILLS

Teaching & Assessment

Assignment/lab design; instructional materials and videos; answer keys and solution notebooks; Kaggle competition hosting; grading and feedback; office hours/forums; autograding; mentoring

Programming & Machine Learning

Python; MATLAB; C++; Java; SQL; NumPy; pandas; SciPy; Matplotlib; scikit-learn; OpenCV; TensorFlow/Keras; PyTorch; Jupyter; TensorBoard

Systems & Cloud

Git; Linux; AWS; Terraform; Docker; CI/CD; PostgreSQL; MongoDB; Cassandra; REST APIs

PROFESSIONAL EXPERIENCE

Cisco

Jan 2023 - Present

Cloud platforms and optical networking | Bengaluru, India

Optical Software Development Engineer II

Jan 2025 - Present

- Designed and implemented a comprehensive unit-testing framework that reduced testing time for new features by more than 90%, improving release velocity and test consistency.
- Developed end-to-end data-plane software for NCS 1014/Aquila optical line cards, including CDR hardware integration, secure boot, and QPSK modulation for greater reach.
- Created internal tools for automated builds and deployments, near-real-time device-log streaming, and performance-monitoring extraction to accelerate debugging and developer workflows.

Software Engineer

Aug 2023 - Jan 2025

- Developed reusable Terraform modules for AWS infrastructure and parallelized deployment plans, cutting deployment time by 50% while improving consistency across services.
- Identified root causes of four production outages and resolved performance bottlenecks across PostgreSQL, MongoDB, and Cassandra, reducing page-load time by 40%.
- Mentored two new engineers through structured knowledge-transfer sessions, code reviews, code-quality expectations, and hands-on debugging.

Software Development Intern

Jan 2023 - Jul 2023

- Built a CI pipeline that generated and published Python and Java client SDKs on API changes and delivered an AWS Glue data-integration job for the PX Cloud platform.

Schneider Electric | Summer Intern

Jun 2022 - Aug 2022

- Independently designed and built a software tool that reduced switchgear test-case discussion time by more than 90%, from two days to two hours.

ACADEMIC AND RESEARCH PROJECTS

- SWIFT - Lightweight Image Super-Resolution:** Co-developed the official PyTorch implementation of a SwinV2 Transformer and Fast Fourier Convolution model, including training, benchmarking, Docker, and TorchServe inference.
- Multi-view 3D Reconstruction:** Implemented an incremental Structure from Motion pipeline for sparse 3D point-cloud reconstruction using SIFT, pose estimation, triangulation, and bundle adjustment.
- Chocollvm Compiler Frontend:** Collaborated on a ChocoPy-subset compiler frontend that produces type-checked ASTs and LLVM IR, with a CLI and automated tests.
- SSML - Spark Streaming for Machine Learning:** Collaborated on distributed CIFAR-10 image classification using Apache Spark/PySpark, TensorFlow/Keras, and Kafka for a Big Data course project.
- Bitcoin Protocol from Scratch:** Built SHA-256, RIPEMD-160, secp256k1 operations, wallet generation, transaction signing, and a Bitcoin testnet transaction without external dependencies.
- Restaurant Analysis using the Yelp Dataset:** Used Yelp check-in and review data for restaurant closure-risk analysis, exploratory analysis, classification, and LDA-based feature discovery.
- Systems and Parallel Programming Coursework:** C and Python assignments covering multithreaded Pi computation, thread-error analysis, and parallel-programming exercises.
- Cloud Provisioning using RDBMS:** Developed a database-backed application for allocating cloud hardware to users and tracking assignments, supported by ER modeling and full-stack components.
- Cloud-Hack: Kubernetes Microservices:** Deployed a containerized Flask/MongoDB blog as Kubernetes microservices using Deployments, Services, ConfigMaps, and Secrets.
- PeTra - Pet Travel and Sitter Booking:** Equal-contribution React application for booking pet-friendly hotels and pet sitters, developed for Web Technologies I.

PUBLICATIONS

- [1] Co-author, "Toward Faster and Efficient Lightweight Image Super-Resolution Using Transformers and Fourier Convolutions," Artificial Intelligence and Applications, 3(2), 168-178, 2025.
- [2] Co-author, "Monitoring and Alert Systems for Underwater Data Centers using Arduino," 2021 IEEE International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), pp. 1-6.

HONORS AND CERTIFICATIONS

AWS Certified Developer Associate	MRD Scholarship PES University	Web Application Security CTF Cisco organization-level winner
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